

YOURSTORY



H2S



COCO CLI

HACKATHON

GCC Edition

Forgotten Follow-ups

Making sure every “repeat this scan” and “refer to a surgeon” actually happens

Team Name : Ruzen Tech

Team Leader Name : Mahaboobsab (Sahil) Makandar, AI Solutions Architect

Team Size : 2 (with Rojina Mallick, Developer)

Problem Statement : Patient and Member 360 and Clinical or Regulatory Document Copilot

Most recommended follow-up scans never happen

37% → 74%

Lung-nodule follow-ups completed before vs after a tracking programme (442 of 1,202, then 465 of 632). Nodule Net, Respiratory Medicine 2022

39% → 68%

Completion at East Alabama Medical Center after adopting tracking software; staff time fell from 5 hours a week to 15 minutes

8 mm → stage 4A

Alleged in a lawsuit reported Sept 2026: a nodule found in 2022 was never followed up; stage 4A lung cancer in 2025 (Roper St. Francis, US)

Who feels it

Hospital quality and radiology leaders, follow-up coordinators, and payer care-management teams (many run from India GCCs).

Why today's tools miss it

- Tracking tools only see their own hospital's records.
- A scan done at another hospital looks like a miss; a real miss looks like nothing.
- Coordinators chase charts by hand, with no proof trail.

What we change

One ranked worklist that joins report text, hospital records and payer claims (shared zero-copy), with the exact source line behind every status.

Every follow-up gets a status, a reason and a source line

1 Read

Cortex AI pulls each “do this, by this date” from the report, with a verbatim quote



2 Rule

Guideline logic in plain SQL: Fleischner 2017, SVS 2018, ACR



3 Check

Hospital orders and reports, plus payer claims through a Secure Data Share



4 Act

Ranked worklist, cited copilot, clinician-approved letter

Red: overdue

Due date passed and nothing matching was found. The overdue Alert fires.

e.g. 9 mm nodule, CT due in 3 months, nothing after 4

Amber: done elsewhere

A payer claim shows the right test at another hospital. The outside report is requested.

e.g. CT chest claim (CPT 71250) from another hospital

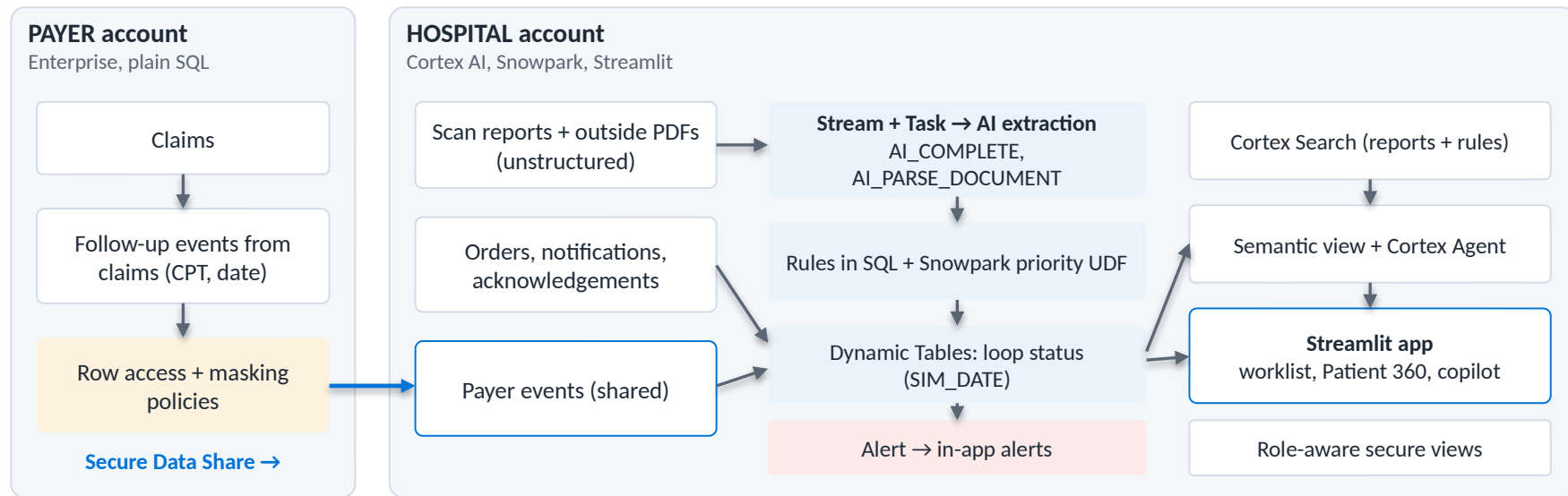
Green: verified

The outside report (even a PDF, read with AI_PARSE_DOCUMENT) is in hospital records and AI_FILTER confirms it discusses the original finding.

e.g. “RUL nodule stable” in the new report

Not a diagnostic tool. It never overrides the radiologist and never contacts a patient without clinician approval. A chest X-ray claim can never close a CT follow-up.

Architecture: two accounts, one share, no copied data



CoCo CLI

builds and runs it

[followup-intake](#)

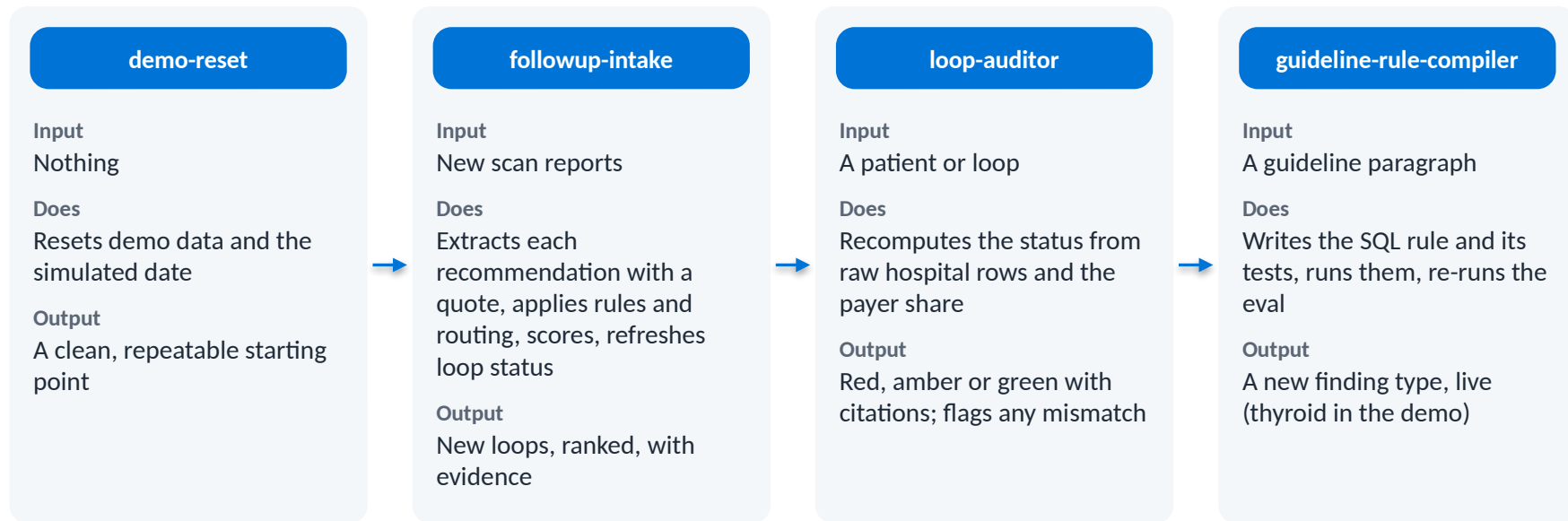
[loop-auditor](#)

[guideline-rule-compiler](#)

[demo-reset](#)

PreToolUse hook guards PHI and access controls. Bundled skills used for semantic view, agent, search, Dynamic Tables and Streamlit.

Four CoCo CLI skills run the workflow end to end



Guardrail: a PreToolUse hook blocks PHI-like strings, dropping access controls and destructive DDL on curated tables. **Evidence:** skills, plans and session logs are committed in `.cortex/` and `docs/coco-log.md`.

Safe by design: AI reads, rules decide, clinicians approve

1 Rules, not guesses

Status comes from deterministic SQL. AI only extracts, quotes and explains.

2 Every claim has a source

Each status shows the verbatim report line, checked word for word in code.

3 Wrong test never closes a loop

A chest X-ray claim (CPT 71045/71046) cannot close a CT follow-up. The reason is shown.

4 Nobody silently dropped

Known cancer, immunosuppression, under 35 and screening patients are rerouted, with the reason.

5 Told and done, tracked apart

Clinician acknowledged and patient notified, separately from the scan being done.

6 Governed data

Payer row access and masking on the share; role-aware views at the hospital; clinician approves every letter.

Red-teamed by us: 41 attacks on the copilot, roles and intake. 27 held, 10 bugs found and fixed, 4 known limits, all listed in [docs/RED_TEAM.md](#).

Measured, not claimed

Test sets

Set A, real text: 120 Indiana University chest X-ray reports, labelled row by row by the team.

Set B, tricky reports: 36 reports full of traps. Answers fixed before the text was generated, then reviewed by a team member with clinical lab experience.

Set C, scale: 2,000 synthetic patients and 843 reports, generated from a hidden answer key written first.

Baselines: a keyword rule, and AI alone with no rules.

Traps in Set B

“No nodule seen”, “stable since 2023”, an addendum that changes the plan, two findings in one report, hedges, known cancer, AAA at the threshold, a 9 x 7 mm nodule, a chest X-ray claim offered as closure.

Test	Ours	Keyword	AI only
Set B: correct of 36	36	18	21
Set B: false greens	0	5	5
Set C: loop accuracy (700)	99.1%	61.4%	60.4%*
Set C: false greens	0	94	13*
Set A: real follow-ups found (of 24)	21†	13	24 (9 false)
Quotes verified	99.9%	-	-
Cost per 1,000 reports	≤ 1.3 credits	-	-
Copilot answers correct (of 15)	15‡	-	-

†Blind run: 17 of 24 (91.7% accuracy); 21 of 24 (95.0%) after one rule fix made after seeing Set A. *AI-only scored on a sample of 96. ‡Native agent evaluation; 11 of 15 before a clinic-filter fix. Latency: 21 ms per report.

Impact: more patients get the scan they were told they need

36% → 79%

Follow-up completion on our synthetic hospital: own records only, vs payer share plus worklist (simulated)

+15 points

Gained from payer claims alone (35.9% → 50.9%), the part single-hospital tools cannot see (simulated)

0

False greens across 736 test reports: no patient was wrongly marked as done

Scales without new infrastructure

- Serverless AI and incremental Dynamic Tables.
- Zero-copy sharing: any number of payers, no data moved.
- New finding types in minutes with guideline-rule-compiler.

Beyond the demo

- More findings: adrenal, liver, pancreatic cysts.
- India: ABDM consent-based records as the bridge between hospitals.
- Native App packaging and regional-language letters.

Who pays: the hospital recovers lost follow-up imaging and lowers malpractice risk; the payer closes care gaps and catches cancers earlier, without handing over its claims history.

Built on Snowflake, end to end

AI

AI_COMPLETE (structured JSON)

AI_PARSE_DOCUMENT

AI_FILTER

Cortex Search

Cortex Agent

Semantic view

Pipeline

Snowpark Python UDF

Stored procedures

Streams and Tasks

Dynamic Tables

Alerts

Notebook (eval harness)

Sharing and governance

Secure Data Sharing

Row access policy

Masking policy

Secure views and roles

Marketplace listing

Resource monitor

Apps and tooling

Streamlit in Snowflake

Worksheets runbook (sql/)

Snowflake CLI

CoCo CLI: 4 skills

CoCo hooks

Everything runs inside Snowflake: no external LLM APIs, servers or vector databases. Python via Snowpark, rules in SQL.

The working app: ranked, cited, measured

On worklist	Red (overdue)	Amber (outside claim)	Patient not told
211	162	49	75
Red and amber loops needing action	Due date passed, no correct test	Done elsewhere, report requested	No notification on record

Finding type	Tier
Choose options	Choose options

	Rank	Patient	Finding	Size	Tier	Status	Days overdue	Priority	Clinician acked	Patient told	Why
☐	1	P00334	CXR_RECOMMEND_CT	15 mm	1	RED	296	3950	☐	☒	Overdue by 296 days
☐	2	P00301	CXR_RECOMMEND_CT	25 mm	1	AMBER		3940	☒	☒	Claim E00301 (71250 at City General Hospita

Ranked worklist: most dangerous first, with who was told

Who is first on the worklist and why?

Loop L-R00334-0 (patient P00334) is first, at worklist rank 1 of 211.

Why it is first:

- Status: RED, tier 1, priority score 3950, overdue by 296 days (due window was 2025-11-12 to 2025-12-12).
- Score breakdown: 3000 (tier 1 band) + 900 time (60% of the due window elapsed) + 50 context (age 65+ and current smoker).
- The worklist is ranked by tier, then priority score, then days overdue, so a tier 1 loop this far past its due window sorts to the top.
- Communication: patient notified = true, but clinician acknowledged = false.

Evidence:

- Report line (R00334, 2025-11-12): "There is an indeterminate opacity in the left lower lobe measuring approximately 15 mm."
- Rule R3 (radiologist recommendation) triggered the follow-up. For a chest X-ray recommending CT, the action is CT chest within 1 month, tier 1 when wording is suspicious or a mass.
- Priority rule R7: Score = tier band (tier 1 = 3000) + share of due window elapsed + context (age 65+, current smoker).

Note: I am not diagnosing or estimating risk here. Any patient letter or referral would be a draft requiring clinician approval before it is sent.

Tools used: explain_priority, search_reports_and_rules

Copilot: cited answer with the report line, rule and score

Reports

▼ R00334 - 2025-11-12 - XR - HOSPITAL

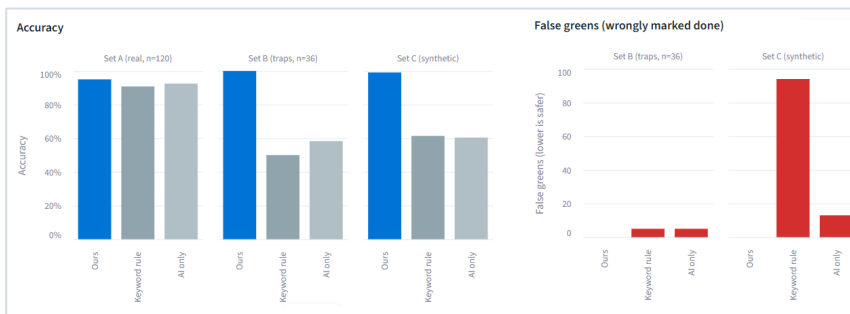
CLINICAL HISTORY
70-year-old male with current smoking history. Chest radiograph obtained for clinical evaluation.

TECHNIQUE
Frontal (PA) and lateral chest radiographs are reviewed.

FINDINGS
There is an indeterminate opacity in the left lower lobe measuring approximately 15 mm. The opacity does not have characteristic benign features. The remainder of the lungs are clear. No pleural effusion or pneumothorax is identified. The cardiac silhouette is normal. The mediastinum is unremarkable. No acute osseous abnormalities are noted.

IMPRESSION
Indeterminate 15 mm opacity in the left lower lobe. Given the patient's smoking history and indeterminate appearance, CT chest with contrast is recommended for further characterization and follow-up evaluation of this finding.

Patient 360: every report, with the evidence quote highlighted



Results: ours vs keyword rule vs AI alone, on all three test sets

YOURSTORY



1125



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THANK YOU

Repo: <https://github.com/sahil-makandar/forgotten-followups>

Live app: https://app.snowflake.com/JVHFISR/pb73401/#/streamlit-apps/FFU.APP.FFU_APP

Judge login: account JVHFISR-PB73401, user JUDGE_RUZEN (password in the submission brief)